Consensus in distributed systems

Consensus is the problem of getting a set of processes to agree on a single value even though some of them may fail. It underpins leader election, atomic commit and replicated state machines.

The FLP result shows that no deterministic protocol can guarantee consensus in an asynchronous system where even one process may crash.

Raft

Raft decomposes consensus into leader election, log replication and safety. A term is a logical clock, and at most one leader is elected per term.

Log entries flow only from the leader to followers, which makes the protocol easier to reason about than Paxos.

Sharding

Sharding partitions data across nodes by a key so that no single node holds the whole dataset. The partitioning function determines how evenly load is spread.

Resharding is the expensive part, because it moves data while the system continues to serve traffic.